

Capgemini - 100 Technical MCQ Questions

Capgemini

For BTech Computer Science / IT Students

Object-Oriented Programming

1. Which of the following is true about a constructor?

- A) It has a return type
- B) It is called automatically when an object is created
- C) It can be inherited like other methods
- D) It must be private

2. Which access modifier makes a member accessible only within the same class?

- A) public
- B) protected
- C) private
- D) default

3. What is encapsulation?

- A) Hiding implementation details and exposing only functionality
- B) Creating multiple classes
- C) Compiling code faster
- D) Using loops effectively

4. Which OOP principle is primarily used to reduce code duplication by reusing existing classes?

- A) Inheritance
- B) Polymorphism
- C) Abstraction
- D) Encapsulation

5. Which concept allows objects of different classes to be treated as objects of a common superclass?

- A) Polymorphism
- B) Encapsulation
- C) Abstraction
- D) Overloading

6. Which keyword is used in Java to prevent a class from being inherited?

- A) static
- B) final
- C) private
- D) abstract

7. What happens if a class does not implement all abstract methods of an interface it implements?

- A) Compilation succeeds silently
- B) The class must be declared abstract
- C) Runtime error occurs
- D) Nothing happens

8. What is a virtual function used for in C++?

- A) To achieve runtime polymorphism
- B) To avoid inheritance
- C) To create static memory
- D) To improve compilation speed

9. What does 'this' keyword refer to in Java/C++?

- A) The parent class
- B) The current object instance
- C) A static variable
- D) The compiler

10. What is the purpose of the 'super' keyword in Java?

- A) To create a static method
- B) To refer to the parent class members
- C) To delete an object
- D) To define an interface

11. Which of these is an example of compile-time polymorphism?

- A) Method overriding
- B) Operator overloading (in C++)
- C) Dynamic method dispatch
- D) Runtime type identification

12. Constructor overloading is possible when:

- A) Constructors have same name but different parameter lists
- B) Constructors are private
- C) A class has no constructor
- D) Only one constructor exists

13. Which of these best defines an interface in Java?

- A) A class with only static methods
- B) A contract of methods a class must implement
- C) A private class
- D) A type of exception

14. An abstract class in Java can have:

- A) Only abstract methods
- B) Only concrete methods
- C) Both abstract and concrete methods
- D) No methods at all

15. What is the main difference between an abstract class and an interface (pre-Java 8)?

- A) Interfaces can have constructors
- B) Abstract classes cannot have any methods
- C) Interfaces cannot have method bodies while abstract classes can
- D) There is no difference

16. A class that cannot be instantiated directly is called a(n):

- A) Abstract class
- B) Final class
- C) Static class
- D) Sealed class

17. In Java, which keyword is used to inherit a class?

- A) implements
- B) extends
- C) inherits
- D) super

18. Which of the following supports multiple inheritance in Java?

- A) Classes
- B) Interfaces
- C) Constructors
- D) Packages

DBMS & SQL

19. Which index type stores data rows in the same order as the index?

- A) Non-clustered index
- B) Clustered index
- C) Bitmap index
- D) Hash index

20. Which of the following commands removes all rows from a table but keeps its structure, and cannot be rolled back in most databases?

- A) DELETE
- B) DROP
- C) TRUNCATE
- D) REMOVE

21. What is a materialized view?

- A) A view whose result set is physically stored and periodically refreshed
- B) A temporary table only
- C) A synonym for an index
- D) A type of trigger

22. What is denormalization?

- A) The process of adding redundancy to improve read performance
- B) Removing all keys from a table
- C) A type of join
- D) Encrypting a database

23. Which type of JOIN returns all rows from the left table and matched rows from the right table?

- A) INNER JOIN
- B) LEFT JOIN
- C) RIGHT JOIN
- D) CROSS JOIN

24. Which of the following best explains 'eventual consistency' in distributed/NoSQL databases?

- A) All nodes are always immediately consistent
- B) Given enough time without new updates, all nodes will converge to the same value
- C) Data is never consistent
- D) It refers only to backup consistency

25. What is a view in SQL?

- A) A physical copy of a table
- B) A virtual table based on the result of a SQL query
- C) A type of index
- D) A stored procedure

26. Which SQL aggregate function returns the number of rows matching a condition?

- A) SUM()
- B) COUNT()
- C) AVG()
- D) MAX()

27. What does ACID stand for in DBMS transactions?

- A) Atomicity, Consistency, Isolation, Durability
- B) Accuracy, Consistency, Integration, Data
- C) Atomic, Complete, Independent, Durable
- D) Association, Consistency, Isolation, Data

28. Which of the following is true about a candidate key?

- A) It can contain NULL values
- B) Every candidate key can be a primary key, but only one is chosen as such
- C) A table can have only one candidate key
- D) It must always be a foreign key

29. Which normal form removes partial dependency on a composite primary key?

- A) 1NF
- B) 2NF
- C) 3NF
- D) BCNF

30. Which SQL command is used to change the structure of an existing table?

- A) UPDATE
- B) ALTER
- C) MODIFY
- D) CHANGE

31. What is a composite key?

- A) A key formed by combining two or more columns to uniquely identify a row
- B) A key that references another table
- C) An automatically generated key
- D) A key with only one column

32. Which SQL clause is used to filter groups after a GROUP BY?

- A) WHERE
- B) HAVING
- C) FILTER
- D) ORDER BY

33. What is a deadlock in the context of a database transaction?

- A) Two or more transactions waiting indefinitely for resources locked by each other
- B) A syntax error in SQL
- C) A missing index
- D) A type of join

34. What does DBMS stand for?

- A) Database Management System
- B) Data Backup Management Service
- C) Database Model Structure
- D) Data Base Modeling Software

35. What is the purpose of an index in a database?

- A) To enforce referential integrity
- B) To speed up data retrieval operations
- C) To store backup data
- D) To encrypt table data

36. What is a primary key?

- A) A column that can contain duplicate values
- B) A column (or set of columns) that uniquely identifies each row and cannot be NULL
- C) An index used only for sorting
- D) A key used only in foreign tables

37. A foreign key is used to:

- A) Uniquely identify a row in its own table
- B) Establish a link between two tables
- C) Encrypt sensitive data
- D) Speed up all queries automatically

38. Which property of a transaction ensures that partial updates are never visible to other transactions?

- A) Atomicity
- B) Consistency
- C) Isolation
- D) Durability

Operating Systems

39. What is the purpose of a context switch?

- A) To save the state of a running process and load the state of another
- B) To terminate a process
- C) To allocate memory
- D) To format a disk

40. What is the purpose of a semaphore?

- A) To manage access to shared resources and avoid race conditions
- B) To compile code
- C) To manage file names
- D) To render graphics

41. What is a race condition?

- A) When multiple processes/threads access shared data concurrently and the outcome depends on timing
- B) A process running faster than expected
- C) A type of scheduling algorithm
- D) A network protocol

42. Which of these describes a 'zombie process'?

- A) A process that has completed execution but still has an entry in the process table
- B) A process that never starts
- C) A process using 100% CPU
- D) A kernel-level thread

43. Which of the following is true about a microkernel architecture?

- A) It moves most services to user space, keeping the kernel minimal
- B) It has no scheduler
- C) It cannot support IPC
- D) It is always slower with no exceptions

44. What is the difference between paging and segmentation?

- A) Paging divides memory into fixed-size blocks; segmentation divides memory into variable-sized logical units
- B) They are identical
- C) Segmentation is faster in all cases
- D) Paging is used only for disk storage

45. What is virtual memory?

- A) Physical RAM only
- B) A memory management technique that gives an application the illusion of a large, contiguous memory using disk space
- C) A type of cache memory
- D) A backup memory device

46. Which of the following best describes multiprogramming?

- A) Running multiple programs on multiple CPUs simultaneously only
- B) Keeping multiple programs in memory so the CPU always has one to execute, improving utilization
- C) Running only one program at a time
- D) A type of scheduling exclusive to real-time systems

47. Which of the following synchronization tools is used to prevent race conditions?

- A) Compiler
- B) Semaphore
- C) Scheduler
- D) Loader

48. What does 'thrashing' refer to in an OS?

- A) Excessive CPU usage by a single process
- B) A state where the system spends most of its time swapping pages instead of executing processes
- C) A type of deadlock
- D) A network congestion issue

49. Which scheduling algorithm gives the shortest average waiting time (in theory, for a given batch)?

- A) FCFS
- B) SJF (Shortest Job First)
- C) Round Robin with a large quantum
- D) Priority Scheduling

50. What is a critical section in OS?

- A) A section of code that accesses shared resources and must not be executed by more than one process/thread at a time
- B) A section reserved for the kernel only
- C) A type of page fault
- D) A memory segment for stack only

51. Which page replacement algorithm suffers from Belady's Anomaly?

- A) LRU
- B) FIFO
- C) Optimal
- D) LFU

52. Which of the following is a preemptive scheduling algorithm?

- A) FCFS
- B) Round Robin
- C) Non-preemptive SJF
- D) Batch scheduling

53. Which of the following is an example of an inter-process communication (IPC) mechanism?

- A) Pipes
- B) Compilers
- C) Loaders
- D) Linkers

54. What is a monolithic kernel?

- A) A kernel where all OS services run in kernel space as a single large process
- B) A kernel with no drivers
- C) A kernel used only in embedded systems
- D) A kernel that cannot be modified

Computer Networks

55. What is the maximum number of usable host addresses in a /24 subnet?

- A) 256
- B) 254
- C) 255
- D) 512

56. What does VPN stand for and what is its main purpose?

- A) Virtual Private Network - creates a secure, encrypted connection over a public network
- B) Virtual Public Network - increases internet speed
- C) Verified Private Node - a type of router
- D) Virtual Protocol Number - an IP addressing scheme

57. Which protocol is used to send emails?

- A) POP3
- B) SMTP
- C) HTTP
- D) FTP

58. What does 'three-way handshake' refer to in TCP?

- A) The SYN, SYN-ACK, ACK process used to establish a connection
- B) A method to terminate a connection
- C) An error-detection mechanism
- D) A DNS lookup process

59. What is a firewall used for?

- A) To speed up internet connections
- B) To monitor and control incoming/outgoing network traffic based on security rules
- C) To assign IP addresses
- D) To translate domain names

60. Which of the following is true about IPv6 compared to IPv4?

- A) IPv6 uses 32-bit addresses while IPv4 uses 128-bit
- B) IPv6 uses 128-bit addresses, providing a vastly larger address space than IPv4's 32-bit addresses
- C) IPv6 is identical to IPv4 in address length
- D) IPv4 has replaced IPv6 entirely

61. Which of the following is a connectionless protocol?

- A) TCP
- B) UDP
- C) FTP
- D) HTTP over TCP

62. What is the purpose of the ARP protocol?

- A) To resolve an IP address to a MAC address
- B) To resolve a domain name to an IP address
- C) To transfer files
- D) To send email

63. What is NAT (Network Address Translation) used for?

- A) To translate private IP addresses to a public IP address for internet access
- B) To encrypt data
- C) To assign MAC addresses
- D) To resolve domain names

64. What is the purpose of a subnet?

- A) To divide a large network into smaller, more manageable segments
- B) To increase the number of MAC addresses
- C) To eliminate the need for routers
- D) To replace DNS

65. Which of the following best describes latency in networking?

- A) The amount of data transferred per second
- B) The delay before data transfer begins following an instruction
- C) The number of devices on a network
- D) The physical distance between two devices

66. Which layer of the OSI model handles encryption and data formatting?

- A) Presentation Layer
- B) Session Layer
- C) Application Layer
- D) Physical Layer

67. Which of the following best describes a subnet mask?

- A) It divides an IP address into network and host portions
- B) It encrypts network traffic
- C) It assigns MAC addresses
- D) It is used only in IPv6

68. What is the function of a DNS server?

- A) To translate human-readable domain names into IP addresses
- B) To assign private IP addresses within a LAN
- C) To encrypt HTTP traffic
- D) To manage email delivery

Data Structures & Algorithms

69. Which data structure uses FIFO (First In First Out) order?

- A) Stack
- B) Queue
- C) Tree
- D) Graph

70. Which data structure is best suited for implementing a priority queue efficiently?

- A) Array
- B) Linked List
- C) Heap
- D) Stack

71. Which traversal of a Binary Search Tree gives elements in sorted order?

- A) Pre-order
- B) Post-order
- C) In-order
- D) Level-order

72. What is the time complexity of binary search on a sorted array of n elements?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(n \log n)$
- D) $O(1)$

73. What is the time complexity of accessing an element in an array by index?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$
- D) $O(n \log n)$

74. What is the time complexity of inserting an element at the head of a singly linked list?

- A) $O(n)$
- B) $O(1)$
- C) $O(\log n)$
- D) $O(n \log n)$

75. What is a balanced binary search tree used to prevent?

- A) Overflow errors
- B) Degeneration into a linked list (worst-case $O(n)$ operations)
- C) Memory leaks
- D) Integer overflow

76. Which of these best describes the 'sliding window' technique?

- A) Maintaining a subset of elements (a window) over a sequence and adjusting its bounds to solve subarray/substring problems efficiently
- B) A sorting technique for linked lists
- C) A hashing technique
- D) A tree traversal method

77. What is the worst-case time complexity of QuickSort?

- A) $O(n \log n)$
- B) $O(n^2)$
- C) $O(\log n)$
- D) $O(n)$

78. What does BFS stand for in graph traversal?

- A) Best First Search
- B) Breadth First Search
- C) Binary First Search
- D) Backtrack First Search

79. Which data structure uses LIFO (Last In First Out) order?

- A) Queue
- B) Stack
- C) Linked List
- D) Array

80. Which data structure is typically used to implement BFS?

- A) Stack
- B) Queue
- C) Heap
- D) Array only

81. Which algorithmic technique does Merge Sort primarily use?

- A) Greedy
- B) Divide and Conquer
- C) Dynamic Programming
- D) Backtracking

82. What is the space complexity of an iterative approach to computing Fibonacci numbers with memoization storing all n results?

- A) $O(1)$
- B) $O(n)$
- C) $O(n^2)$
- D) $O(\log n)$

83. What is tail recursion?

- A) A recursive call that is the last operation in a function, which can be optimized by some compilers into iteration
- B) A recursive function with no base case
- C) A function that calls itself twice
- D) A type of infinite loop

84. Which sorting algorithm is considered stable and has $O(n \log n)$ time complexity in all cases?

- A) QuickSort
- B) Merge Sort
- C) Selection Sort
- D) Heap Sort

Programming Language Fundamentals

85. In Python, which of the following is a mutable data type?

- A) Tuple
- B) String
- C) List
- D) int

86. In Java, what is autoboxing?

- A) Automatic conversion between primitive types and their corresponding wrapper classes
- B) A way to box multiple objects together
- C) A garbage collection technique
- D) A type of casting error

87. What does 'pass by reference' mean?

- A) A copy of the variable's value is passed to the function
- B) The function receives a reference to the actual variable, so changes affect the original
- C) Variables cannot be passed to functions
- D) Only primitive types can be passed

88. In C, what does the 'malloc' function do?

- A) Deallocates memory
- B) Allocates a block of memory dynamically
- C) Initializes a variable to zero
- D) Declares a static variable

89. What is a memory leak?

- A) When allocated memory is never freed even though it is no longer needed, wasting resources over time
- B) When a program uses too little memory
- C) A type of compilation error
- D) A CPU scheduling issue

90. What is the default value of an uninitialized boolean instance variable in Java?

- A) true
- B) false
- C) null
- D) 0

91. What does 'immutable' mean for an object (e.g., a String in Java)?

- A) Its value can be changed after creation
- B) Its state cannot be changed after it is created
- C) It can only be used once
- D) It cannot be passed to functions

92. Which keyword is used to handle exceptions in Python?

- A) catch
- B) except
- C) rescue
- D) handle

93. Which of the following is a value type (not a reference type) in Java?

- A) int
- B) String
- C) Array
- D) Object

94. What is the purpose of the 'finally' block in exception handling (Java/Python)?

- A) It executes only if an exception occurs
- B) It executes only if no exception occurs
- C) It always executes regardless of whether an exception occurred or not
- D) It is used to declare exceptions

Web & Cloud Fundamentals

95. Which HTTP method is typically used to retrieve data without side effects?

- A) POST
- B) GET
- C) DELETE
- D) PUT

96. What does HTTP stand for?

- A) HyperText Transfer Protocol
- B) HighText Transmission Protocol
- C) Hyperlink Transfer Text Protocol
- D) HyperText Trade Protocol

97. What is the purpose of a Content Delivery Network (CDN)?

- A) To cache and serve content from servers geographically closer to users, reducing latency
- B) To encrypt all web traffic
- C) To replace DNS servers
- D) To manage database transactions

98. What is the primary difference between HTTP and HTTPS?

- A) HTTPS uses encryption (SSL/TLS) to secure data in transit while HTTP does not
- B) HTTP is faster in all cases
- C) HTTPS cannot use port 443
- D) There is no difference

99. What does 'statelessness' mean in the context of REST APIs?

- A) The server stores client session state between requests
- B) Each request from a client contains all the information needed to process it, with no reliance on stored server-side session state
- C) The API has no state at all, including no data
- D) Stateless APIs cannot use authentication

100. What is a RESTful API?

- A) An API that strictly requires SOAP protocol
- B) An architectural style for designing networked applications using stateless operations over standard HTTP methods
- C) A type of database
- D) A programming language

Answer Key

1-B	2-C	3-A	4-A	5-A	6-B	7-B	8-A	9-B	10-B
11-B	12-A	13-B	14-C	15-C	16-A	17-B	18-B	19-B	20-C
21-A	22-A	23-B	24-B	25-B	26-B	27-A	28-B	29-B	30-B
31-A	32-B	33-A	34-A	35-B	36-B	37-B	38-C	39-A	40-A
41-A	42-A	43-A	44-A	45-B	46-B	47-B	48-B	49-B	50-A
51-B	52-B	53-A	54-A	55-B	56-A	57-B	58-A	59-B	60-B
61-B	62-A	63-A	64-A	65-B	66-A	67-A	68-A	69-B	70-C
71-C	72-B	73-C	74-B	75-B	76-A	77-B	78-B	79-B	80-B
81-B	82-B	83-A	84-B	85-C	86-A	87-B	88-B	89-A	90-B
91-B	92-B	93-A	94-C	95-B	96-A	97-A	98-A	99-B	100-B

Format: [Question No.]-[Correct Option Letter]. Content compiled from publicly available 2026 placement-prep resources and established CS-fundamentals question banks - verify current-year specifics before your interview.